Recognition Programs Improve Performance: A Meta-Analysis of 61 Samples

SYNTHETIC TEST PAPER, NOT REAL RESEARCH

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(All authors, primary studies, firms, and data sets in this paper are fictional.)

Abstract

Prior studies disagree about whether workplace recognition programs pay off, and the disagreement is sharpest in small firms. We meta-analyze 180 effect sizes from 61 samples (N = 45,960) and find that recognition programs improve unit and firm performance (corrected rho = .12). The effect is consistently positive across settings, which resolves the inconsistency in prior findings. Our second main finding is that the effect holds even in small firms. A meta-analytic structural equation model traces the effect through employee engagement, voluntary turnover, and customer satisfaction. Moderator analyses identify two boundary conditions that lie in program design: manager training and public delivery.

Keywords: employee recognition, firm performance, meta-analysis, MASEM

1 Introduction

Firms spend heavily on programs that recognize employees: awards, public thanks, service milestones, and small non-cash rewards. Whether these programs are associated with better unit and firm performance is still debated. Some studies of recognition and return on assets (ROA) report clear positive associations (see the Cordero and Hollis 1997 and Zeller and Kaur 2014 rows of Table 1), while others report associations near zero (the Pacheco 1994 and Valcourt 2009 rows). The disagreement is sharpest in small firms, where the few available studies point in different directions and where managers have the least slack to spend on programs that may not pay off.

This paper pools the evidence. We meta-analyze 180 effect sizes from 61 samples of units and firms, covering published and unpublished work from 1990 to 2024. The paper makes three contributions. First, and most important, it shows that the relationship between recognition and performance generalizes across settings, including small firms, where prior findings disagreed; this resolves the inconsistent findings of earlier work. Second, it uses meta-analytic structural equation modeling (MASEM) to trace the pathway that links recognition to ROA through employee engagement, voluntary turnover, and customer satisfaction. Third, it identifies boundary conditions of the relationship that lie in how programs are designed and delivered.

2 Theory and Hypotheses

Recognition signals to employees that their effort is seen and valued. Social exchange accounts hold that employees who feel valued reciprocate with engagement and discretionary effort, and that engaged employees are less likely to quit. Lower turnover preserves firm-specific knowledge

and service relationships, which should show up in customer satisfaction and, in the end, in ROA. These arguments lead to three hypotheses.

Hypothesis 1. Recognition program intensity is positively related to unit and firm ROA.
Hypothesis 2. Employee engagement, voluntary turnover, and customer satisfaction link recognition program intensity to ROA.
Hypothesis 3. The relationship between recognition and ROA is stronger when managers are trained to give recognition and when recognition is delivered in public.

3 Method

3.1 Literature search

We searched ABI/INFORM Global, Business Source Complete, APA PsycInfo, the Web of Science Core Collection, ProQuest Dissertations and Theses Global, SSRN, and the Academy of Management Proceedings for work dated from January 1990 through December 2024. The last search ran on 20 December 2024. Appendix A gives the full search string used in each database. The dissertation database, SSRN, and the Academy of Management Proceedings were included to find unpublished work: dissertations, working papers, and conference papers. Records from all sources are counted together in Figure 1. Section 4.4 compares published and unpublished effect sizes.

3.2 Inclusion rules and screening

Before coding began, we wrote down the inclusion rules and the coding manual. A study was eligible if it (a) reported on a sample of employees, units, or firms, (b) measured the presence or intensity of a recognition program, (c) measured unit or firm ROA, and (d) reported a correlation between the two or a statistic that can be converted to one. Two authors screened every title and abstract independently and agreed on 94 percent of screening decisions; they resolved disagreements by discussion. Figure 1 shows the flow of records in the PRISMA 2020 format (Page et al. 2021).

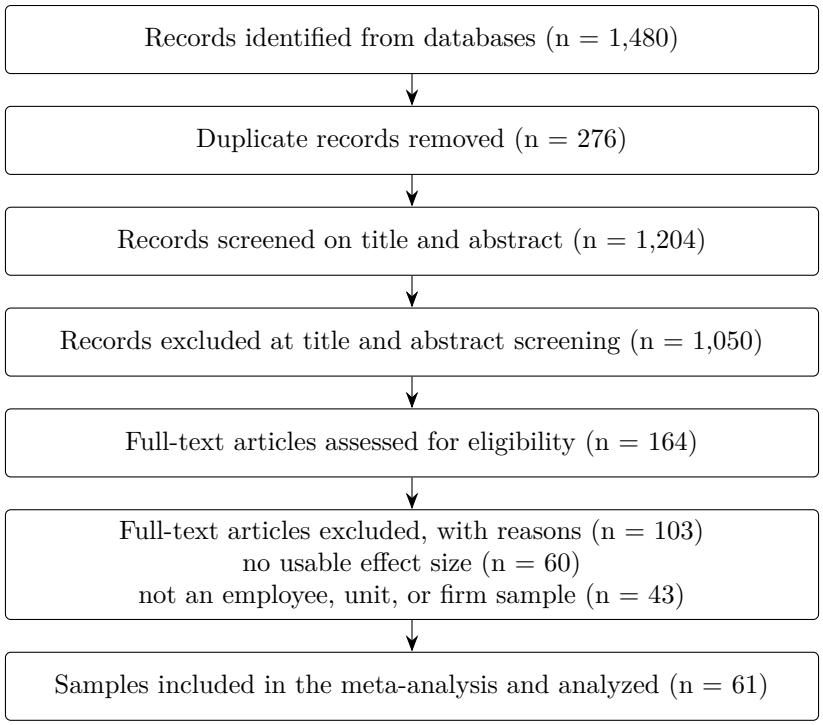


Figure 1: Flow of records through the search and screening. Each included article reports one sample.

3.3 Coding

Two authors coded every sample independently. Their agreement on effect sizes and sample sizes was 96 percent, and Cohen's kappa for the categorical codes was .89; disagreements were resolved by discussion. We coded every reported relationship between a recognition measure and ROA, whether or not the primary study hypothesized it. Effect sizes were aligned so that a positive value means that more recognition goes with higher ROA; the two samples that measured the absence of recognition had their signs reversed. A sample gave several effect sizes when it reported several recognition measures or several years of ROA. Table 1 lists the 61 samples.

Of the 180 effect sizes, 142 are zero-order correlations, reported directly or converted from a standardized mean difference (Appendix B) or a bivariate t statistic (Appendix C). The other 38 of the 180 effect sizes are partial correlations from regressions of ROA on recognition with control variables. We converted each regression t statistic to a partial correlation (Appendix C) and pooled these with the zero-order correlations.

Samples measured ROA at the unit level (for example, stores or plants) or at the firm level; we coded the level of analysis and test it as a moderator. We also coded reward form, region, the timing of the ROA measure, program formality, manager training, and delivery. Manager training and delivery were often not reported. Three included articles draw on the National Workplace Panel.[1]

[1]The National Workplace Panel is fictional. It was invented for this synthetic test paper, and no real data set is described.

3.4 Analysis

We converted each correlation to Fisher's z and pooled the 180 independent effect sizes in a standard random-effects model, estimated by restricted maximum likelihood (REML; Veroniki et al. 2016), in the Hedges and Olkin tradition. Results are transformed back to the correlation metric for reporting. Confidence intervals use the normal distribution. We report heterogeneity as Q, I-squared, and tau on the Fisher z scale. I-squared is computed from Q and its degrees of freedom as (Q minus df) divided by Q.

3.5 Correction for measurement error

We also report an estimate corrected for measurement error in the recognition measure, following the psychometric tradition of Schmidt and Hunter (2015). We label it rho to set it apart from the random-effects estimates, which are uncorrected. Few primary studies reported the reliability of their recognition measure, so we assumed a reliability of .70, a typical value, for every recognition measure, and a reliability of 1.00 for ROA. The corrected estimate is the uncorrected mean divided by the square root of the product of the two reliabilities.

Table 1: Included samples

Study	Status	Industry	Size	Data source	Data years	N	ES
Abernathy and Ncube (1999)	Pub	Services	Large	Company records	1998	1,223	4
Achterberg and Faye (2009)	Pub	Services	Small	Own survey	2003-2006	86	4
Adeyemi (2010)	Unpub	Services	Large	Records and survey	2008-2009	1,261	6
Albrecht and Ncube (2022)	Pub	Services	Large	Records and survey	2020	646	3
Amundsen and Aziz (2021)	Pub	Services	Large	Own survey	2020	685	4
Arvidsson and Wolff (2010)	Pub	Services	Large	Own survey	2005-2009	377	3
Bakshi and Tan (2019)	Pub	Services	Large	Own survey	2014-2018	1,184	4
Barros and Mensah (1998)	Pub	Services	Small	Company records	1992-1995	63	3
Belcourt and Wolff (2000)	Pub	Services	Small	Records and survey	1995-1998	146	3
Birkeland (2024)	Pub	Services	Large	Own survey	2021	915	2
Bramwell and Kaur (1999)	Pub	Services	Large	Company records	1998	954	4
Brandvold et al. (2019)	Pub	Services	Large	Company records	2013-2017	6,434	4
Chaudhry and Mukherjee (1998)	Unpub	Services	Large	Own survey	1993-1997	838	5
Cordero and Hollis (1997)	Pub	Services	Small	Own survey	1993-1995	48	3
Corrigan and Kaur (2007)	Pub	Services	Small	Records and survey	2002-2005	71	3
Dalgaard (2005)	Pub	Manufacturing	Large	Field experiment	2002	523	1
Delacroix (1996)	Pub	Manufacturing	Large	Own survey	1990-1994	410	1
Drummond and Park (2018)	Pub	Manufacturing	Large	Records and survey	2015	580	1
Duarte (2019)	Pub	Public sector	Large	Company records	2017	413	1
Eberhardt (2009)	Pub	Services	Small	Company records	2007	188	3
Engstrom and Tan (1997)	Unpub	Manufacturing	Large	Records and survey	1991-1995	305	1
Fairweather and Tan (2024)	Pub	Services	Large	Own survey	2023	762	5
Farouk and Mensah (2006)	Pub	Services	Large	Own survey	2002-2004	1,301	6
Fenwick (2000)	Unpub	Manufacturing	Large	Company records	1995-1998	276	1
Gallardo and Mukherjee (1996)	Pub	Services	Large	Own survey	1993	569	4
Gunnarsson (1994)	Pub	Manufacturing	Large	Company records	1989-1991	212	1
Hakkarainen and Wolff (2000)	Pub	Services	Small	Own survey	1999	132	4

Continued on next page

Table 1, continued

Study	Status	Industry	Size	Data source	Data years	N	ES
Halloran and Mbeki (2015)	Pub	Services	Large	National Workplace Panel	2009-2014	2,412	3
Holmgren (1997)	Pub	Services	Large	Company records	1994	416	5
Ibarra and Park (2013)	Unpub	Public sector	Large	Records and survey	2008-2011	812	1
Iversen and Quaye (2016)	Pub	Services	Large	National Workplace Panel	2009-2014	2,412	4
Jaramillo and Sethi (2012)	Pub	Manufacturing	Large	Own survey	2011	388	1
Jovanovic and Ferreira (2002)	Pub	Services	Small	Company records	2000-2001	163	3
Kallio and Hollis (2009)	Pub	Services	Large	Own survey	2003-2007	454	4
Kowalczyk (2017)	Pub	Services	Large	National Workplace Panel	2009-2014	2,412	2
Laurent (2020)	Pub	Services	Large	Own survey	2015-2019	531	2
Lindahl and Park (2008)	Pub	Services	Large	Own survey	2001-2005	1,069	3
Madsen (2016)	Pub	Services	Small	Own survey	2012-2014	94	3
Marchetti and Ferreira (2016)	Pub	Public sector	Large	Records and survey	2012-2015	688	1
Nakagawa and Brennan (2011)	Unpub	Services	Large	Own survey	2009	1,107	4
Nieminen and Ferreira (2006)	Unpub	Services	Large	Company records	2001-2005	1,146	6
Oduya and Hollis (2022)	Pub	Services	Large	Records and survey	2020	992	5
Okafor and Lind (2016)	Pub	Services	Large	Company records	2010-2014	960	1
Pacheco (1994)	Pub	Services	Small	Own survey	1991	108	4
Petrosyan and Brennan (2008)	Pub	Manufacturing	Large	Records and survey	2002-2006	354	1
Quintero and Faye (2013)	Pub	Services	Large	Own survey	2010-2011	723	4
Rasmussen (2012)	Pub	Public sector	Large	Company records	2007-2009	515	1
Rautio and Aziz (1999)	Unpub	Public sector	Large	Own survey	1997	745	2
Sandoval (2006)	Pub	Manufacturing	Large	Company records	2003-2005	447	1
Sasaki and Ncube (2009)	Pub	Services	Large	Own survey	2004-2008	608	5
Takahashi (2009)	Unpub	Manufacturing	Large	Records and survey	2007-2008	489	1
Thorvaldsen and Mukherjee (1996)	Unpub	Services	Small	Company records	1995	57	4
Uzoma and Brennan (2009)	Unpub	Public sector	Large	Records and survey	2008	602	2
Valcourt (2009)	Unpub	Services	Small	Company records	2007-2008	117	3
Vantongeren and Ruiz (2018)	Pub	Manufacturing	Large	Field experiment	2016	100	1
Varga (1998)	Pub	Services	Large	Records and survey	1992-1996	877	4
Westergaard and Aziz (2017)	Pub	Services	Large	Records and survey	2016	493	5
Whitcombe and Mensah (2014)	Pub	Services	Large	Company records	2011	800	3
Yilmaz and Faye (2008)	Pub	Manufacturing	Large	Records and survey	2003-2005	331	1
Zapata (2009)	Pub	Public sector	Large	Own survey	2003-2007	905	2
Zeller and Kaur (2014)	Pub	Services	Large	Records and survey	2011	1,031	4

Note. All studies in this table are fictional; they were invented for this synthetic test paper and have no entries in the reference list. Pub = published; Unpub = unpublished (dissertation, working paper, or conference paper). Small = fewer than 100 employees. N = units or firms. ES = number of effect sizes coded from the sample. Total N = 45,960; total ES = 180.

4 Results

4.1 Overall relationship

Table 2 reports the pooled estimates. Across all samples, the mean correlation between recognition and ROA is r = .10 (95 percent CI .07 to .13; k = 180), which supports Hypothesis 1. Heterogeneity is substantial: Q(179) = 1,125.0, I-squared = 84.1 percent, and tau = .12 on the Fisher z scale. Corrected for measurement error in the recognition measure, rho = .12. The relationship is positive across settings: every industry and firm size subgroup in Table 3 has a positive mean correlation.

Table 2: Pooled relationship between recognition and ROA

	k	N	Estimate	95% CI	Q	df	I^2 (%)	τ
All samples, r	180	45,960	.10	[.07, .13]	1,125.0	179	84.1	.12
Without the largest sample, r	176	39,526	.09	[.06, .12]	1,080.0	175	83.8	.12
All samples, corrected rho	180	45,960	.12	[.08, .16]				

Note. k = number of effect sizes. r = random-effects mean (REML), transformed back from Fisher's z. τ is on the Fisher z scale. The corrected rho divides r and its interval by the square root of .70 times 1.00 (Schmidt and Hunter 2015). The largest sample is the Brandvold et al. (2019) row of Table 1.

4.2 Settings

Table 3 reports the relationship by industry and firm size. In manufacturing, each of the 12 samples contributed one effect size, so the 12 effect sizes come from 12 independent samples. The mean is r = .20 (95 percent CI .16 to .24), with tau = .04 on the Fisher z scale. The 95 percent prediction interval, computed with a t distribution on k minus 2 degrees of freedom, runs from .10 to .29, so the relationship is positive across the manufacturing settings studied.

Small firms matter most for the debate that motivates this paper. Pooling the 40 effect sizes from 12 small-firm samples gives r = .05 (95 percent CI .01 to .09). The interval excludes zero, so the effect holds even in small firms, the setting in which earlier studies disagreed most.

Table 3: Recognition and ROA by industry and firm size

Subgroup	k	Samples	N	r	95% CI	Q	df	I^2 (%)	τ
Industry									
Manufacturing	12	12	4,415	.20	[.16, .24]	15.6	11	29.5	.04
Services	158	42	36,865	.09	[.06, .12]	1,015.0	157	84.5	.12
Public sector	10	7	4,680	.08	[−.02, .18]	40.0	9	85.0	.14
Firm size									
Small firms	40	12	1,273	.05	[.01, .09]	63.0	39	38.1	.08
Large firms	140	49	44,687	.11	[.08, .14]	905.0	139	84.6	.12

Note. k = number of effect sizes. Each subgroup is a separate random-effects model (REML). τ is on the Fisher z scale. Small firms have fewer than 100 employees.

4.3 Sensitivity to influential samples

The largest sample (the Brandvold et al. 2019 row of Table 1; N = 6,434) carries 14 percent of total N. Without it, the mean is r = .09 (95 percent CI .06 to .12; k = 176), as Table 2 reports. Leave-one-out analyses, which drop one sample and all of its effect sizes at a time, give means from .09 to .10. Following Viechtbauer and Cheung (2010), we also computed studentized deleted residuals and Cook's distances for every sample; no studentized deleted residual exceeded 3 in absolute value, and no Cook's distance stood apart from the rest.

4.4 Publication bias

The fail-safe N is 1,340: that many unpublished null results would be needed to bring the mean correlation to nonsignificance, far more than the 5k + 10 = 910 threshold. We conclude that

publication bias is not a threat to these results. Trim and fill (Duval and Tweedie 2000) imputed 14 effect sizes on the left side of the funnel plot; the adjusted mean, r = .08 (95 percent CI .05 to .11), is the bias-free estimate, and it is still positive. Published and unpublished work give similar results: r = .11 (95 percent CI .08 to .14; k = 144) for published samples and r = .07 (95 percent CI .02 to .12; k = 36) for unpublished samples. A test of the difference between the two subgroups is not significant (Q-between(1) = 1.81, p = .18).

4.5 Moderators

We tested seven moderators one at a time by estimating the relationship separately at each level of the moderator (Table 4). Five moderators, including the level of analysis, show a significant positive relationship at every level. Two do not. When managers were trained to give recognition, r = .12 (p = .04; k = 9); when they were not, r = .02 (p = .80; k = 6). When recognition was delivered in public, r = .12 (p = .04; k = 11); when it was private, r = .03 (p = .65; k = 8). Manager training and public delivery are therefore confirmed boundary conditions of the recognition and performance relationship, which supports Hypothesis 3.

Table 4: Moderator analyses

Moderator	Level	k	r	95% CI	p
Level of analysis	Unit	104	.11	[.09, .13]	< .001
	Firm	76	.09	[.06, .12]	< .001
Reward form	Monetary	70	.10	[.07, .13]	< .001
	Non-monetary	110	.10	[.08, .12]	< .001
Region	North America	96	.10	[.07, .13]	< .001
	Europe	52	.11	[.07, .14]	< .001
	Asia	32	.09	[.04, .13]	< .001
Timing of ROA	Lagged	64	.09	[.06, .12]	< .001
	Same year	116	.11	[.09, .13]	< .001
Program formality	Formal	150	.10	[.08, .12]	< .001
	Informal	30	.09	[.04, .14]	< .001
Manager training	Trained	9	.12	[.01, .24]	.04
	Not trained	6	.02	[−.13, .17]	.80
Delivery	Public	11	.12	[.01, .23]	.04
	Private	8	.03	[−.10, .16]	.65

Note. k = number of effect sizes. Each row is a separate random-effects model (REML). p tests whether the subgroup mean differs from zero. Manager training was not reported for 165 effect sizes and delivery for 161.

4.6 Meta-analytic structural equation model

To test Hypothesis 2, we built a pooled correlation matrix for five constructs: recognition, employee engagement, voluntary turnover, customer satisfaction, and ROA. We pooled each of the ten correlations separately with a random-effects model, using every included sample that reported that correlation and averaging within a sample that reported several. Many primary studies reported correlations only among the variables in their own models, so the cells rest on different sets of samples: cell k runs from 3 to 60 and cell N from 900 to 45,000 (Table 5). No cell is empty. The MASEM uses zero-order correlations only; one sample (Okafor and Lind 2016) reported only a

partial correlation between recognition and ROA, so the recognition and ROA cell has k = 60.

We fit a path model in which recognition predicts engagement and turnover, engagement predicts turnover and customer satisfaction, and engagement, turnover, and customer satisfaction predict ROA, following the correlation-pooling approach described by Bergh et al. (2016). We fit the model with N = 12,000, the average cell sample size. The model fits well: chi-square(3) = 60.4, RMSEA = .040, and CFI = .99. Table 5 reports the standardized path coefficients.

Table 5: MASEM: pooled correlations and path model

Panel A. Pooled correlations					*Panel B. Standardized paths*	
Pair	k	N	r	τ	Path	Coefficient
Recognition, ROA	60	45,000	.10	.12	Recognition to engagement	.30
Recognition, engagement	41	36,000	.30	.10	Recognition to turnover	−.05
Engagement, ROA	33	12,000	.17	.09	Engagement to turnover	−.32
Recognition, turnover	25	9,000	−.15	.08	Engagement to satisfaction	.25
Turnover, ROA	19	6,000	−.13	.07	Engagement to ROA	.10
Engagement, turnover	14	4,200	−.34	.08	Turnover to ROA	−.08
Recognition, satisfaction	11	3,000	.13	.06	Satisfaction to ROA	.18
Satisfaction, ROA	8	2,400	.21	.07		
Engagement, satisfaction	5	1,500	.25	.05		
Turnover, satisfaction	3	900	−.09	.04		

Note. k = number of samples in the cell; N = total N in the cell. Each correlation is a separate random-effects mean (REML); τ is on the Fisher z scale. Satisfaction = customer satisfaction; turnover = voluntary turnover.

5 Discussion

Recognition programs are among the most common and least expensive tools managers have, yet the evidence on whether they pay off has been mixed. Pooling 61 samples, we find a positive relationship between recognition and ROA that is consistently positive in every industry and firm size we examined, including small firms. The inconsistency in earlier findings is resolved: individual studies disagreed because they were small, not because the relationship differs across settings.

The MASEM establishes the mechanism. Recognition raises engagement; engaged employees are less likely to leave and serve customers better; and lower turnover and higher customer satisfaction in turn raise ROA. The moderator results add that recognition works when managers are trained to give it and when it is given in public.

The size of the relationship is modest. Across the included samples, the standard deviation of ROA averages 4.0 percentage points, so units or firms one standard deviation higher in recognition intensity have ROA about 0.4 percentage points higher (.10 times 4.0). For a firm with 1 billion dollars in assets, that difference is 4 million dollars a year in operating return.

For managers, the results imply that recognition programs, especially those in which managers are trained and recognition is public, are a sound investment in firms of every size. For researchers, the results suggest that further single-sample studies of whether recognition matters add little; attention should turn to how programs are designed.

Our study has limits. It covers ROA only, so it says nothing about other measures of performance, and it covers work through December 2024. Most samples come from North America

and Europe.

Data availability

The coded data are available from the authors on request.

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Appendix A. Search strings

Each database was searched from January 1990 through December 2024, with no language restriction. The core string was the same in every database; field codes differed.

ABI/INFORM Global and ProQuest Dissertations and Theses Global (abstract, title, and subject fields). noft(("recognition program*" OR "employee recognition" OR "reward and recognition" OR "service award*" OR "employee of the month") AND ("return on assets" OR ROA OR "financial performance" OR "unit performance" OR "firm performance"))

Business Source Complete (TI, AB, and SU fields). ("recognition program*" OR "employee recognition" OR "reward and recognition" OR "service award*" OR "employee of the month") AND ("return on assets" OR ROA OR "financial performance" OR "unit performance" OR "firm performance")

APA PsycInfo (title, abstract, and keywords). the core string above.

Web of Science Core Collection. TS=(("recognition program*" OR "employee recognition" OR "reward and recognition" OR "service award*" OR "employee of the month") AND ("return on assets" OR ROA OR "financial performance" OR "unit performance" OR "firm performance"))

SSRN and Academy of Management Proceedings (title, abstract, and keywords). the core string above, entered one recognition term at a time because these sites do not accept nested Boolean strings.

Appendix B. Converting standardized mean differences

Two included samples were field experiments that compared units that received a recognition program with comparison units and reported a standardized mean difference, d, for ROA. Because the groups were unequal in size, we converted d to r with

$$r = \frac{d}{\sqrt{d^2 + a}}, \qquad a = \frac{(n_1 + n_2)^2}{n_1 n_2},$$

where n1 and n2 are the two group sizes. As a worked case, Vantongeren and Ruiz (2018) report d = .50 for 30 recognized stores and 70 comparison stores. Then a = 100 squared divided by 2,100, which is 4.762, and r = .50 divided by the square root of 5.012, which gives r = .223.

Appendix C. Converting t statistics

Four samples reported only a t test of the bivariate relationship between recognition and ROA. We converted each to a correlation with

$$r = \frac{t}{\sqrt{t^2 + N}},$$

where N is the sample size. For example, a bivariate t = 2.50 in a sample of N = 100 converts to r = .243.

For the 38 partial correlations, we converted the t statistic on the recognition coefficient in each regression with the residual degrees of freedom of that regression:

$$r_{\text{partial}} = \frac{t}{\sqrt{t^2 + df_{\text{residual}}}}.$$